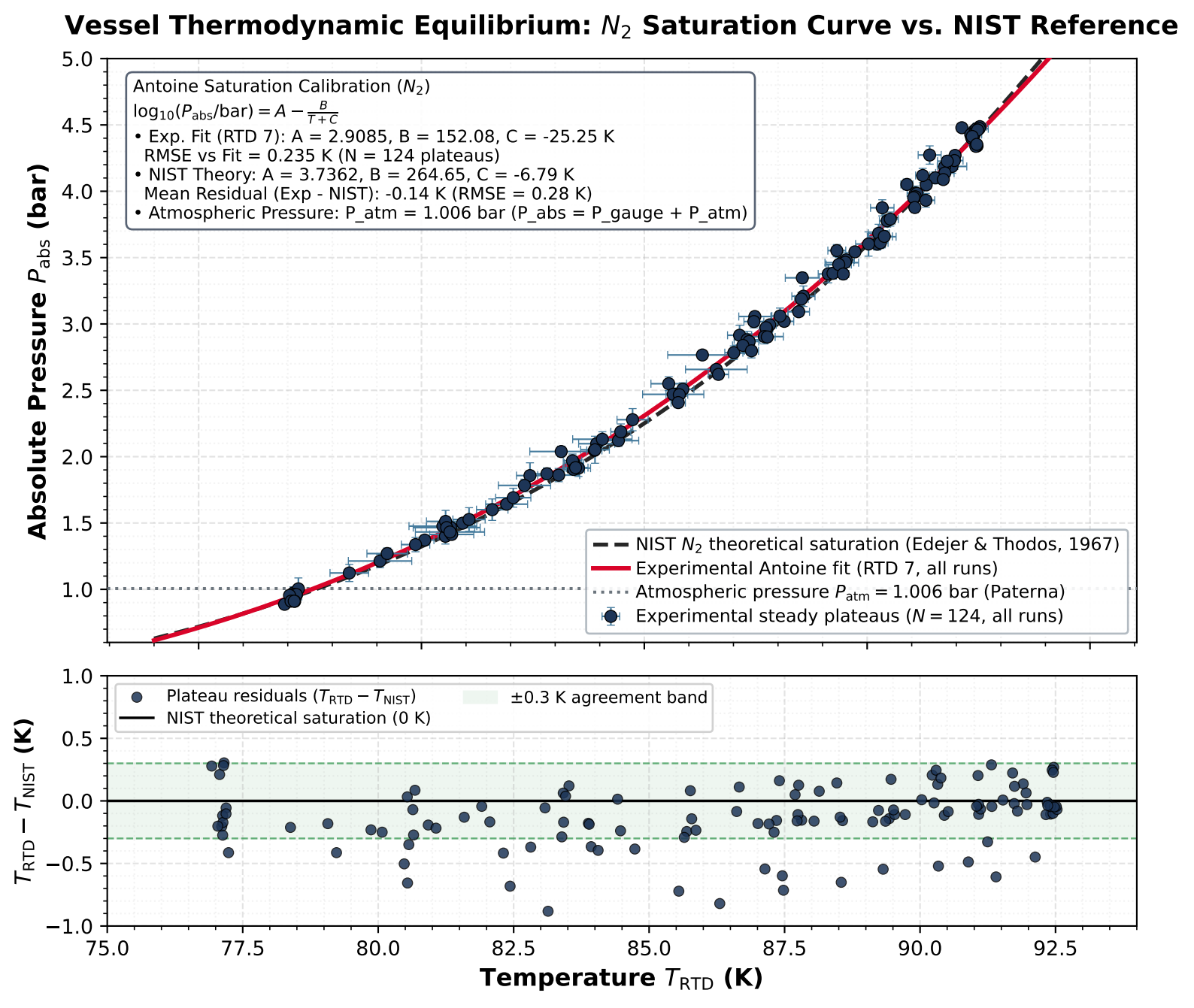

Vessel Thermodynamic Equilibrium: N_2 Saturation Curve vs. NIST Reference
Antoine Saturation Calibration (N_2)
$\log_{10}(P_{abs}/\text{bar}) = A - \frac{B}{T+C}$
• Exp. Fit (RTD 7): A = 2.9085, B = 152.08, C = -25.25 K
RMSE vs Fit = 0.235 K (N = 124 plateaus)
• NIST Theory: A = 3.7362, B = 264.65, C = -6.79 K
Mean Residual (Exp - NIST): -0.14 K (RMSE = 0.28 K)
• Atmospheric Pressure: P_atm = 1.006 bar (P_abs = P_gauge + P_atm)
NIST N_2 theoretical saturation (Edejer & Thodos, 1967)
Experimental Antoine fit (RTD 7, all runs)
Atmospheric pressure $P_{atm} = 1.006$ bar (Paterna)
Experimental steady plateaus ($N = 124$, all runs)
Absolute Pressure P_{abs} (bar)
Plateau residuals ($T_{RTD} - T_{NIST}$)
NIST theoretical saturation (0 K)
±0.3 K agreement band
$T_{RTD} - T_{NIST}$ (K)
Temperature T_{RTD} (K)